

**PIMPRI-CHINCHWAD EDUCATION TRUST'S
S.B.PATIL COLLEGE OF SCIENCE AND COMMERCE**

**Std. 12th
Marks - 70**

**Prelim Exam-2
BIOLOGY**

**Date -31 /03/21
Time - 3hrs.**

General Instructions:

- 1) Section A: Given section contains Ten multiple choice type of questions carrying One mark each. All question are compulsory.
Q.No.1 to Q.No.8 contains Eight very short answer type of questions carrying One mark each. All question are compulsory
- 2) Section B: Given section contains Twelve short answer type of questions carrying Two marks each. **Attempt Any Eight Questions.**
- 3) Section C: Given section contains Twelve short answer type of questions carrying Three marks each. **Attempt Any Eight Questions.**
- 4) Section D: Given section contains five long answer type of questions carrying Four marks each. **Attempt Any Three Questions.**
- 5) Neat diagrams must be drawn wherever necessary.

Section A

1) Multiple Choice questions

10

- i) Interferons are produced against _____ infection
a) Viral b) Bacterial c) Fungal d) Protozoal
- ii) Geitonogamy involves
a) Fertilization of a flower by the pollen from a flower of another plant belonging to a distant population
b) Fertilization of a flower by the pollen from another flower of the same plant
c) Fertilization of a flower by the pollen from the same flower
d) Fertilization of a flower by the pollen from a flower of another plant in the same population
- iii) The vas deferens receives duct from the seminal vesicle and opens into urethra as:
a) Epididymis b) Efferent ductile
c) Ejaculatory duct d) Ureter
- iv) Which fungi is used in production of Vitamin C
a) *Saccharomyces cerevisiae* b) *Lactobacillus*
c) *Claviceps* d) *Aspergillus niger*
- v) The mammalian Graafian follicle produces
a) Oestrogen b) Progesterone c) Prolactin d) Oxytocin
- vi) Which of the following statements concerning Griffith's work with *Streptococcus pneumonia* is incorrect?
a) Injected heat-killed S-form bacteria allowed mice to survive
b) Injected S-form bacteria killed mice
c) Injected R-form bacteria killed mice
d) Injected heat-killed S-form bacteria together with live R-form bacteria killed mice
- vii) The transcribing enzyme is:
a) Ligase b) DNA polymerase
c) RNA polymerase d) amino-acyl transferase
- viii) Arteries shows presence of
a) Bicuspid valve b) Tricuspid valve

- ix) _____ is the connecting link between ape and man
 c) Semilunar valve d) No valve
 a) Ramapithecus b) Australopithecus
 c) Homo habilis d) Homo erectus
 x) Amebiasis is a _____ disease
 a) Viral b) Bacterial c) Helminthic d) Protozoal

2) Very Short Answer type questions

8

- i) what is bio fortification?
- ii) Which part in humans is common passage for food and air?
- iii) Give an example of multiple allelism
- iv) What are palindromes?
- v) Define gene pool
- vi) Give one example of plant growth hormone cytokinin
- vii) Which organism is used by space travellers as source of proteins?
- viii) Define population.

Section B

• Short Answer type questions (2 Marker) (Any 8)

16

- 3) Draw a neat labelled diagram of human sperm
- 4) Write a note on endosperm
- 5) What are vestigial organs? Give 2 examples of vestigial organs in Humans
- 6) Explain the process of Biogas production.
- 7) What is the difference between Pulmonary and Systemic circulation?
- 8) Write a note on RBCs in humans
- 9) Mention about causative agents of Malaria, Typhoid, Ring worm, Ascariasis
- 10) Write a note on Cranial nerves. Mention names of any 2 cranial nerves
- 11) What is the difference between endocrine and exocrine glands with examples?
- 12) Write a short note on Restriction Endonucleases
- 13) What is humulin?
- 14) What is mutualism? Explain with an example.

Section C

• Short answer type questions (3 marker) (Any 8)

24

- 15) Draw a neat labelled diagram of VS of anatropous ovule
- 16) Write a note on Lac operon

- 17) Explain Oogenesis
- 18) Write a short note on different types of chromosomes according to their shape.
- 19) Write a short note on Acquired immunity.
- 20) Explain pollination in details.
- 21) Write a short note on posterior pituitary gland
- 22) Explain different types of Biodiversity?
- 23) Explain structure of Neuron with neat labelled diagram
- 24 Define Bio patent, Bio piracy and Bioethics
- 25) Explain double fertilization in angiosperms
- 26) Explain different respiratory volumes.

Section D

• **Long answer type questions (Any 3)**

12

- 27) Explain life cycle of Human Immuno deficiency Virus
- 28) What are X linked diseases. Why the inheritance in X linked disorders called criss-cross inheritance
- 29) Explain process of double circulation with a neat labelled diagram.
- 30) Explain any 4 properties of DNA
- 31) Write a note on conduction of nerve impulse with a neat labelled diagram